

$$S_n = \frac{a(r^n - 1)}{r - 1}, |r| > 1$$

• Infinite geometric Series

→ A series of the form $a + ar + ar^2 + \dots$ is called infinite geometric series.
Sum to n terms of geometric series

$$S_n = \frac{a(1 - r^n)}{1 - r} \text{ if } |r| < 1$$

$$\text{As } n \rightarrow \infty, r^n \rightarrow 0$$

$$\therefore S_{\infty} = \frac{a}{1 - r}, |r| < 1$$

Note: S_{∞} exists only when $|r| < 1$

• Arithmetico geometric Series

If $a, a+d, a+2d, \dots$ are in A.P.
and $1, r, r^2, \dots$ are in G.P then
the series of the form

$a \cdot 1 + (a+d)r + (a+2d)r^2 + \dots$ is called
arithmetico geometric series.

e.g. $1 \cdot 1 + 3r + 5r^2 + 7r^3 + \dots$

Exercise - 5.2

① Write with reason which of the following infinite series have sum,

a) $1 + 3 + 3^2 + 3^3 + \dots$

→ Soln,

$$r = \frac{t_2}{t_1} = \frac{3}{1} = 3 > 1$$

Since $|r| > 1$, so ~~it has~~ This series have not sum.

b) $1 + \frac{1}{3} + \frac{1}{3^2} + \frac{1}{3^3} + \dots$

→ Soln,

$$r = \frac{t_2}{t_1} = \frac{\frac{1}{3}}{1} = \frac{1}{3} \times 1 = \frac{1}{3} < 1$$

since $|r| < 1$, so this series have sum. 2/

c) ~~2 + 4 + 8 + \dots~~

→ Soln,

$$r = \frac{t_2}{t_1} = \frac{4}{2} = 2 > 1$$

since $|r| > 1$, so This series doesn't have sum.

d) $0.5 + 0.05 + 0.005 + \dots$

→ Soln,

$$r = \frac{t_2}{t_1} = \frac{0.05}{0.5} = \frac{1}{10} < 1$$

Since $|r| < 1$ so ~~this~~ This series have sum ~~is~~.

2. Find the sum of the infinite series.

$$a) \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots + \frac{1}{8} + \frac{1}{8} + \dots$$

→ Soln,

$$a = \frac{1}{2}$$

$$r = \frac{t_2}{t_1} = \frac{\frac{1}{4}}{\frac{1}{2}} = \frac{1}{4} \times \frac{2}{1} = \frac{1}{2} < 1$$

Since, $|r| < 1$ so This series have sum.

Now,

$$S_{\infty} = \frac{a}{1-r}$$

$$= \frac{\frac{1}{2}}{1 - \frac{1}{2}}$$

$$= \frac{\frac{1}{2}}{\frac{1}{2}}$$

$$= 1$$

$$= 1$$

$$= 1$$

$$\therefore S_{\infty} = 1$$

u.

Q) Prove that $4^{1/3} \cdot 4^{1/9} \cdot 4^{1/27} \dots = 2$.

→ Soln,

$$4^{(1/3 + 1/9 + 1/27 + \dots)} = 4^{1/2}$$

$$\frac{1}{3} + \frac{1}{9} + \frac{1}{27} + \dots = \frac{1}{2}$$

$$a = \frac{1}{3}$$

$$r = \frac{1}{9} = \frac{1}{9} \times \frac{3}{1} = \frac{1}{3} < 1, \text{ sum exists.}$$

$$\frac{1}{3}$$

Now,

$$S_{\infty} = \frac{a}{1-r}$$

$$= \frac{1}{3}$$

$$\frac{1}{1 - \frac{1}{3}}$$

$$= \frac{1}{\frac{2}{3}}$$

$$\frac{2}{3}$$

$$= \frac{1}{2} = \text{P.H.S}$$

proved

Q.E.D.

5.

a) The sum to infinity of a geometric series is 9 and first term is 6. Find the common ratio.

→ Soln,

$$S_{\infty} = 9$$

$$a = 6.$$

$$r = ?$$

we know that,

$$S_{\infty} = \frac{a}{1-r}$$

$$9 = \frac{6}{1-r}$$

$$\text{or, } 1-r = \frac{6}{9} = \frac{2}{3}$$

$$\text{or, } r = 1 - \frac{2}{3}$$

$$\text{or, } r = \frac{1}{3}$$

$$\therefore r = \frac{1}{3}$$

b) The sum to an infinite number of terms of G.S is 20 and the sum of their squares is 50. Find the series.

→ Soln, let, G.S be $a + ar + ar^2 + \dots$

$$S_{\infty} = 20$$

$$\text{i.e. } a + ar + ar^2 + \dots = 20 \rightarrow \textcircled{i}$$

$$a^2 + (ar)^2 + (ar^2)^2 + \dots = 50 \rightarrow \textcircled{ii}$$

From (i)

$$S_{\infty} = \frac{a}{1-r} \Rightarrow \frac{a}{1-r} = 20$$

$$20(1-r) = a \rightarrow (iii)$$

From (ii)

$$S_{\infty} = \frac{a}{1-r}$$

$$\Rightarrow 50 = \frac{a^2}{1-r^2}$$

$$\Rightarrow 50 = \frac{a}{(1-r)} \cdot \frac{a}{(1+r)}$$

$$\Rightarrow 50 = 20 \cdot \frac{a}{1+r}$$

$$\Rightarrow \frac{50}{20} = \frac{a}{1+r}$$

$$\Rightarrow \frac{a}{1+r} = \frac{5}{2}$$

$$\Rightarrow a = \frac{5(1+r)}{2} \rightarrow (iv)$$

From (iii) & (iv)

$$20(1-r) = \frac{5(1+r)}{2}$$

$$40(1-r) = 5 + 5r$$

$$40 - 40r = 5 + 5r$$

$$45r = 35$$

$$r = \frac{35}{45} = \frac{7}{9}$$

Putting the value of a in eqn (iii)

$$20 \left(1 - \frac{7}{9}\right) = a$$

$$\text{or, } 20 \times \frac{2}{9} = a$$

$$\text{or, } \frac{40}{9} = a$$

$$\therefore a = \frac{40}{9}$$

So, Given Series is.

$$\frac{40}{9} + \frac{280}{81} + \frac{1960}{729} + \dots$$

$$2.b) 3^{-1} + 3^{-2} + 3^{-3} + \dots$$

→ Soln,

$$a = 3^{-1} = \frac{1}{3}$$

$$r = \frac{t_2}{t_1} = \frac{3^{-2}}{3^{-1}} = 3^{-2+1} = 3^{-1} = \frac{1}{3} < 1$$

So, Sum exists

$$S_{\infty} = \frac{a}{1-r}$$

$$= \frac{1}{3}$$

$$1 - \frac{1}{3}$$

$$= \frac{1}{3} \div \frac{2}{3} = \frac{1}{2}$$

c) $2 + \sqrt{2} + 1 + \dots$

→ Soln,

$a = 2$

$r = \frac{t_2}{t_1} = \frac{\sqrt{2}}{2} = \frac{\sqrt{2}}{(\sqrt{2})^2} = \frac{1}{\sqrt{2}} < 1$ so

Sum exists.

$S_{\infty} = \frac{a}{1-r}$

$= \frac{2}{1 - \frac{1}{\sqrt{2}}}$

$= \frac{2}{\frac{\sqrt{2}-1}{\sqrt{2}}}$

$= \frac{2}{\sqrt{2}-1}$

$= \frac{2}{1} \times \frac{\sqrt{2}}{\sqrt{2}-1}$

$= \frac{2\sqrt{2}}{\sqrt{2}-1} \times \frac{\sqrt{2}+1}{\sqrt{2}+1}$

$= \frac{2\sqrt{2}(\sqrt{2}+1)}{2-1}$

$= \frac{2(2+\sqrt{2})}{1}$

$= 2(2+\sqrt{2})$

d) $1 - \frac{1}{3} + \frac{1}{9} - \frac{1}{27} + \dots$

→ Soln,

$a = 1$

$r = \frac{t_2}{t_1} = \frac{-1/3}{1} = -\frac{1}{3} < 1$ so sum exists.

$$\begin{aligned}
 S_{\infty} &= \frac{a}{1-r} \\
 &= \frac{1}{1 - \left(-\frac{1}{3}\right)} \\
 &= \frac{1}{1 + \frac{1}{3}} \\
 &= \frac{1}{\frac{4}{3}} \\
 &= \frac{3}{4} \quad \#.
 \end{aligned}$$

c) $6.4 + 0.04 + 0.004 + \dots$

→ Soln,

$$a = 6.4$$

$$r = \frac{t_2}{t_1} = \frac{0.04}{0.4} = 0.1 < 1 \text{ so sum exists,}$$

$$\begin{aligned}
 S_{\infty} &= \frac{a}{1-r} \\
 &= \frac{6.4}{1-0.1} \\
 &= \frac{6.4}{0.9} \\
 &= \frac{64}{9} \quad \#.
 \end{aligned}$$

f) $\frac{a}{H} + \frac{b}{H^2} + \frac{a}{H^3} + \frac{b}{H^4} + \dots$ to ∞ ($|H| > 1$).

→ Soln,

Given series is,

$$\left(\frac{a}{H} + \frac{a}{H^3} + \dots \right) + \left(\frac{b}{H^2} + \frac{b}{H^4} + \dots \right)$$

$$= \frac{\frac{a}{H}}{1 - \frac{1}{H^2}} + \frac{\frac{b}{H^2}}{1 - \frac{1}{H^2}}$$

$$= \frac{\frac{a}{H} + \frac{b}{H^2}}{1 - \frac{1}{H^2}}$$

$$= \frac{aH + b}{H^2 - 1}$$

$$= \frac{aH + b}{H^2 - 1}$$

$$= \frac{aH + b}{H^2 - 1}$$

4.

b) If $x = \frac{1}{1+y}$; $y > 0$, then prove that

$$x + x^2 + x^3 + \dots \text{ to } \infty = y^{-1}$$

→ Soln,

$$x + x^2 + x^3 + \dots \text{ to } \infty$$

$$a = x$$

$$r = \frac{t_2}{t_1} = \frac{n^2}{n} = n$$

$$S_{\infty} = \frac{a}{1-r}$$

$$= \frac{n}{1-n}$$

$$= \frac{1}{1+y}$$

$$= \frac{1+y}{1-1}$$

$$= \frac{1}{1+y}$$

$$= \frac{1+y-1}{1+y}$$

$$= \frac{1}{y}$$

$$= y^{-1} = R.H.S$$

Proved
=

5.c) The sum of the first two term of an infinite G.S is 5 and each term is equal to thrice the sum of all is succeeding terms. Find the sum of infinity of the series.

→ soln,

Let Geometric Series be.

$$a + ar + ar^2 + \dots$$

By question,

$$a + ar = 5$$

$$a(1+r) = 5$$

$$a = \frac{5}{1+r} \rightarrow (i)$$

$$a = 3(ar + ar^2 + \dots)$$

$$a = 3 \times \frac{ar}{1-r}$$

$$\Rightarrow 1-r = \frac{3r}{1}$$

$$\Rightarrow 4r = 1$$

$$\Rightarrow r = \frac{1}{4}$$

Now,

$$a = \frac{5}{1+\frac{1}{4}}$$

$$= \frac{5}{\frac{5}{4}}$$

$$= \frac{5}{1} \times \frac{4}{5}$$

$$= 4$$

Last,

$$S_{\infty} = \frac{a}{1-r}$$

$$= \frac{4}{1-\frac{1}{4}}$$

$$= \frac{4}{1} \times \frac{4}{3}$$

$$= \frac{16}{3}$$

d) Find the first term of a G.P. whose second term is 2 and the sum to infinity is 8.

→ Soln,

$$t_2 = 2$$

$$S_{\infty} = 8$$

let first term of series be a .

$$r = \frac{t_2}{t_1} = \frac{2}{a}$$

$$S_{\infty} = \frac{a}{1-r}$$

$$8 = \frac{a}{1 - \frac{2}{a}}$$

$$\Rightarrow 8 = \frac{a}{\frac{a-2}{a}}$$

$$\Rightarrow 8 = \frac{a}{1} \times \frac{a}{a-2}$$

$$\Rightarrow 8 = \frac{a^2}{a-2}$$

$$\Rightarrow a^2 - 8a + 16 = 0$$

$$\Rightarrow \cancel{a^2} - 4a - 4a + 16 = 0 \quad \text{or} \quad a^2 - 2 \cdot 4 \cdot a + 4^2 = 0$$

$$\Rightarrow (a-4)^2 = 0$$

$$\Rightarrow a-4=0$$

$$\Rightarrow a=4$$

$$\therefore a=4$$

- ⑥ A rubber ball is dropped from a height of 24 feet. At each rebound, it rises to a height which is $\frac{2}{3}$ of ~~the~~ height of the previous fall. If it continues to fall and rebound, ~~it~~ ~~rises~~ in this way, how far will it travel before it finally comes to rest?

$$\begin{aligned} \rightarrow \text{Distance travelled by ball in } \cancel{\text{upper}} \text{ lower direction } (S_1) &= 24 + \frac{2}{3} \times 24 + 24 \times \frac{2}{3} \times \frac{2}{3} + \dots \\ &= \frac{24}{1 - \frac{2}{3}} \\ &= \frac{24}{\frac{1}{3}} \\ &= 24 \times 3 \\ &= 72 \text{ ft.} \end{aligned}$$

$$\begin{aligned} \text{Total distance travelled by ball in upper direction } (S_2) &= 24 \times \frac{2}{3} + 24 \times \frac{2}{3} \times \frac{2}{3} + \dots \\ &= \frac{24 \times \frac{2}{3}}{1 - \frac{2}{3}} \\ &= \frac{48}{3} \times \frac{3}{1} \\ &= 48 \text{ ft} \end{aligned}$$

∴ Total distance covered by ball = $S_1 + S_2$
 $= 72 + 48$ ft
 $= 120$ ft.

- ⑦ The side of a square is each 16 cm. A second square is inscribed by joining the mid-points of the sides successively. A third square is drawn inside the second square in the same way and this process is continued indefinitely many times. Find the sum of the perimeters of all the squares.

→ Soln,

$P_1 = 4l = 4 \times 16 = 64$ cm

For P_2

$l = \sqrt{8^2 + 8^2} = \sqrt{64 + 64}$
 $= 8\sqrt{2}$

So, Series is,

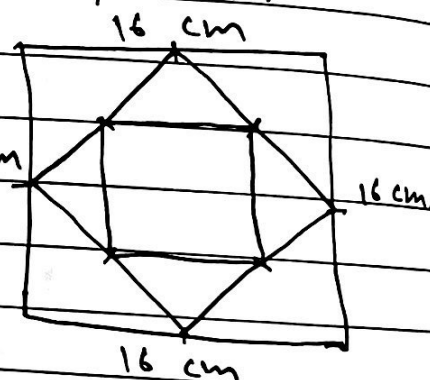
$64 + 8\sqrt{2} + \dots$

$a = 64$

$r = \frac{8\sqrt{2}}{64} = \frac{\sqrt{2}}{8}$

$S_{\infty} = \frac{a}{1-r}$

$= \frac{64}{1 - \frac{\sqrt{2}}{8}}$



$$\begin{aligned}
 &= \frac{64}{\frac{8-\sqrt{2}}{8}} \\
 &= 64 \times \frac{8}{8-\sqrt{2}} \\
 &= \frac{512}{8-\sqrt{2}} \times \frac{8+\sqrt{2}}{8+\sqrt{2}} \\
 &= \frac{512(8+\sqrt{2})}{64-2} \\
 &= \frac{512(8+\sqrt{2})}{62}
 \end{aligned}$$

3. Sum to Infinity of the following Series:

a) $1 + 3x + 5x^2 + 7x^3 + \dots$ to ∞ ($|x| < 1$)

$\therefore$ let $S_{\infty} = 1 + 3x + 5x^2 + 7x^3 + \dots$

$xS_{\infty} = x + 3x^2 + 5x^3 + \dots$

Subtracting, we get

$$\begin{aligned}
 S_{\infty} - xS_{\infty} &= 1 + 3x + 5x^2 + 7x^3 + \dots \\
 &\quad - x - 3x^2 - 5x^3 - 7x^4 - \dots \\
 \hline
 &= 1 + 2x + 2x^2 + 2x^3 + \dots
 \end{aligned}$$

$$\begin{aligned}
 (1-x)S_{\infty} &= 1 + (2x + 2x^2 + 2x^3 + \dots) \\
 &= 1 + \frac{2x}{1-x} = \frac{1-x+2x}{1-x} = \frac{1+x}{1-x}
 \end{aligned}$$

$$\therefore S_{\infty} = \frac{1+x}{(1-x)^2}$$

b) $1 - 5x + 9x^2 - 13x^3 + \dots$ to ∞ ($|x| < 1$)

→ Soln,

let $S_{\infty} = 1 - 5x + 9x^2 - 13x^3 + \dots$

$xS_{\infty} = x - 5x^2 + 9x^3 - 13x^4 + \dots$

Subtracting, we get.

$$S_{\infty} + xS_{\infty} = 1 - 5x + 9x^2 - 13x^3 + \dots$$

$$-x + 5x^2 - 9x^3 + \dots$$

$$1 - 4x + 4x^2 - 4x^3 + \dots$$

$$S_{\infty}(1+x) = 1 - (4x - 4x^2 + 4x^3 - \dots)$$

$$= 1 - \left(\frac{4x}{1 - (-x)} \right)$$

$$= 1 - \frac{4x}{1+x}$$

$$= \frac{1+x-4x}{1+x}$$

$$= \frac{1-3x}{1+x}$$

$$\therefore S_{\infty} = \frac{1-3x}{(1+x)^2}$$